Do Now · retrieval
Debugging and error correction

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Write −1 in 8-bit two's complement.

Q2

What is the last index in a four-item array?

Q3

What precondition does binary search need?

Q4

What happens if code reads a[3] from a three-element C# array?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Check valid indexes — A three-element array has valid indexes 0, 1 and 2.

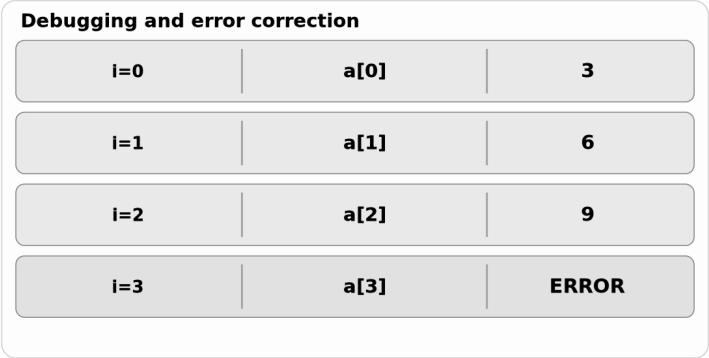
Core idea 2

Follow the condition — When i is 3, i<=a.Length remains true and attempts to read a[3], causing an exception.

Core idea 3

Fix and retest — Use i<a.Length, which stops before i becomes an invalid index.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

off-by-one error — Loop repeats once too many or few times.

exception — Runtime failure or signal raised by an invalid operation.

trace — Record changing execution state.

precondition — Input requirement an algorithm assumes.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

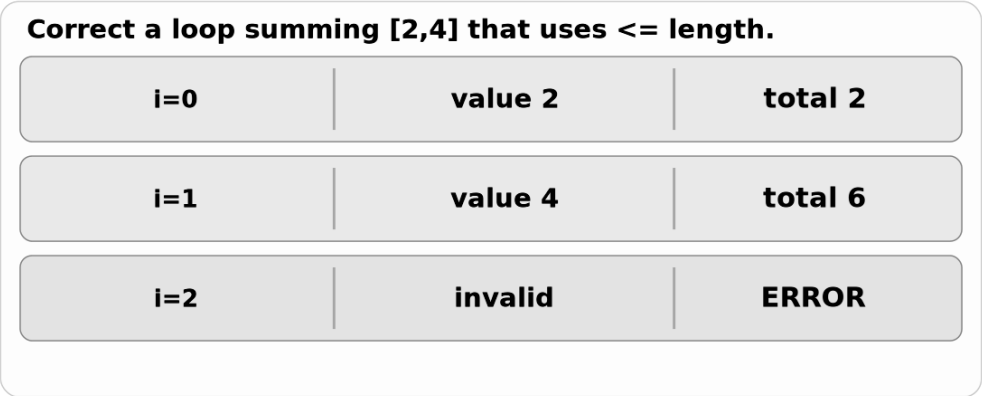
Problem

Correct a loop summing [2,4] that uses <= length.

Follow the worked method

1. Predict — The array's valid indexes are 0 and 1, not 2.
2. Explain — Use i < values.Length and the total is 6.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

With low=3 and high=4, midpoint remains 3 and low=mid fails to shrink the interval.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Fix the loop condition i<=a.Length when visiting every array item.

Question 2

When a[mid]>target, what should binary search do to high?

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Choose normal, boundary and erroneous indexes when a method accepts only indexes 0–4.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Using low=mid+1 after a[mid]<target can prevent non-shrinking binary-search intervals.

Use key vocabulary from memory

1. Explain off-by-one error in your own words.

2. Explain exception in your own words.

3. Explain trace in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.